BPL-700 Magnet Circuit

Spacer height sweep, coil geometry, and power-supply requirements

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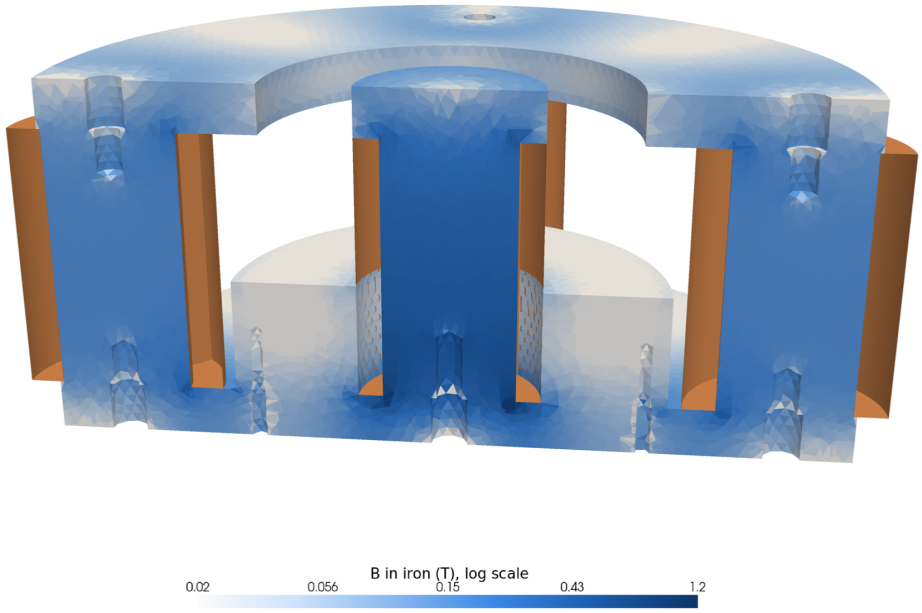


Figure 1: BPL-700 magnet circuit, half-section through the thrust axis and one outer core, baseline geometry (spacer 0.875 in), at the 300 G operating point. Iron is coloured by $|B|$; copper is the five windings. The exit (top plate) is at the top. Nonlinear solve, A36-proxy iron.

Summary

This report sizes the `chamber_spacer`, the A36 ring between the chamber's anode end and the bottom plate. Its height is set by `emag_height` (spacer = `emag_height` − 1.125 in), which is also the height of every coil and core, so a shorter spacer means shorter windings. The inner winding cannot grow outward (the chamber bore sits on it), only inward by thinning the core. 25 full-3D magnetostatic solves cover spacer heights 0–0.875 in in 0.125 in steps at three inner-core diameters (1.000, 0.875, 0.750 in), each scaled to exactly 300 G radial field at the channel exit. It reports the resulting coil geometries, windings and supply requirements; choosing among them is left to the reader.

- **The spacer height hardly changes the field per ampere-turn.** Across the whole sweep the inner coil needs 326–339 A-turns (outer coils 207–215 each) for 300 G, a 4% spread, of which about ±1% is mesh noise. The circuit is gap-dominated, and the exit gap does not move with the spacer.
- **It changes the winding a great deal.** Shorter coils hold fewer turns, so the same A-turns need more current. At the current 1.000 in core, AWG 20 goes from 1.18 A / 3.2 W / 49°C at the full 0.875 in spacer

to 2.04 A / 5.8 W / 72°C with no spacer.

- **Thinning the core buys that room back almost for free.** A 0.750 in core costs about 2% more A-turns but nearly doubles the inner winding window. *No spacer with a 0.750 in core needs 1.30 A and 4.8 W on the inner channel, against 1.18 A and 3.2 W for today's full spacer.*
- **Nothing approaches saturation.** The highest core flux density anywhere in the sweep is 0.74 T (0.750 in core, 0.875 in spacer), against a knee of 1.3–1.5 T on the measured 1020 curve. Core flux *rises* with spacer height: the steel spacer wraps the inner coil and pulls leakage flux through the core.
- **Shorter spacers shape the field slightly better.** The field at the anode end drops from 43 G to 22 G (1.000 in core) as the spacer goes from 0.875 in to 0, closer to the target lens profile, at a cost of about one point of radial purity (97.0% to 95.9%).
- **Supply.** Two independent constant-current channels — one for the inner coil, one for the four outer coils in series. With AWG 20 every swept design runs at 0.62–2.04 A and under 3.1 V hot (under 5.8 W total), so any two-channel supply rated ≥2.5 A and ≥6 V covers them all; a commodity 30 V/3 A unit has ample margin. Other gauges trade current for voltage at the same power (Sec. 5).

1 Configuration and constraints

The circuit is one inner pole and four outer poles between two 0.375 in iron plates (Fig. 2). The top plate is the exit pole piece; the inner core carries a 1.375 in pole cap flush with it, and the radial gap between that cap and the top-plate bore is what puts the field across the discharge channel. Flux returns through the four outer cores and the bottom plate. Inner and outer windings are driven with opposite polarity, on two independent constant-current channels.

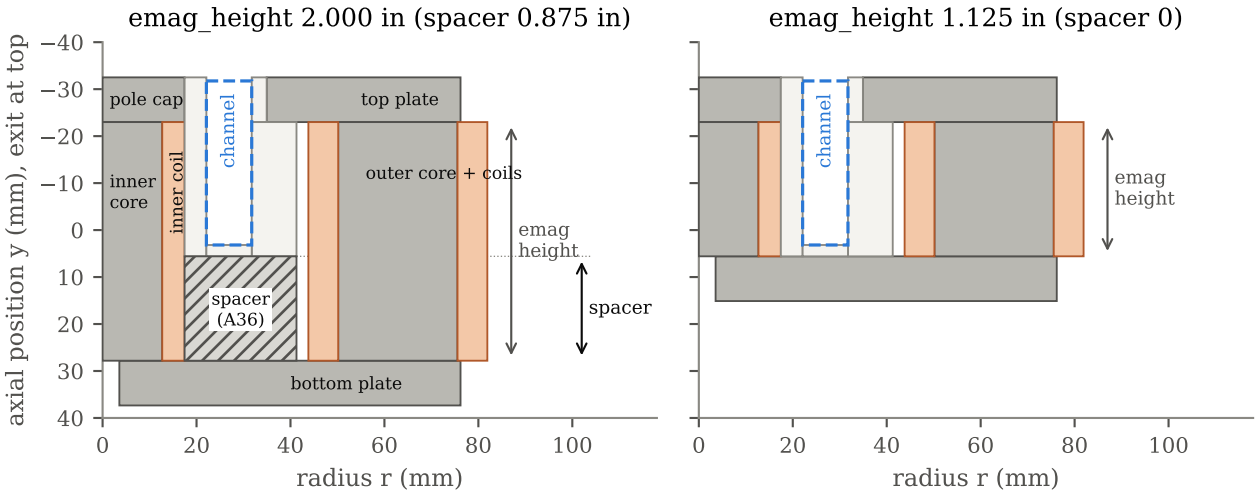


Figure 2: Half-section in the plane of one outer core, at the two ends of the sweep. Grey: iron (A36 plates and spacer; A108 cores). Hatched: `chamber_spacer`. Orange: windings. Light: chamber (PTFE). Dashed blue: the plasma channel used for every field statistic in this report (r 22.10–31.75 mm, anode at $y = +3.18$ mm, exit at $y = -31.75$ mm).

Measured directly from the Onshape STEP exports, not assumed:

- **The spacer is fixed by `emag_height`.** The chamber and top plate never move; the chamber's anode end sits at $y = +5.59$ mm and `emag_height` moves only the bottom plate. So

$$h_{\text{spacer}} = \texttt{emag_height} - 1.125 \text{ in},$$

  and $h_{\text{spacer}} \geq 0$ means `emag_height` ≥ 1.125 in. At exactly 1.125 in Onshape drops the spacer body and the bottom plate sits on the chamber. All coils and cores are `emag_height` tall.
- **The inner coil cannot grow outward.** The chamber bore and the spacer bore are both $r = 17.46$ mm, exactly the inner coil's outer radius (OD 1.375 in). With the chamber fixed, the inner winding window is 4.76 mm × `emag_height` of bare geometry at the current 1.000 in core, or 4.61 mm of usable build once

ground insulation over the core is allowed for. It can only be recovered by thinning the core under the winding (`inner_coil_id`); the pole cap stays at 1.375 in regardless.

- **The outer coils have room.** Outer core axes sit 62.86 mm off-axis and the chamber's outer wall is at $r = 41.28$ mm, so an outer coil can reach ≈ 1.70 in OD before touching it (currently 1.5 in).
- **Stock thicknesses.** The spacer is a single laser-cut A36 part only at a stocked thickness: 0.187, 0.250, 0.313, 0.375 or 0.500 in. Those are marked † in the tables.

2 Materials

Part	Material	Source	Magnetic model
Top, bottom plates; spacer	ASTM A36, hot-rolled P&O	SendCutSend	AISI 1008 curve as proxy (secondhand)
Inner, outer emag cores	ASTM A108 cold-finished bar (taken as 1018)	McMaster	measured 1020 curve, Fermilab TM-1197
Windings	AWG 20 heavy-build polyimide Cu magnet wire, NEMA MW 16-C, 240°C	Remington	$\mu_r = 1$; current density only
Chamber	PTFE	–	$\mu_r = 1$ (not meshed)

Table 1: Materials. Full properties, sources and confidence ratings are in `MATERIALS.md`.

The circuit is gap-dominated, so the channel field barely depends on the iron grade (Sec. 3). Grade does set where a thinned core saturates, which is why the cores are judged against a measured curve (Fig. 7, right). Cold drawing lowers permeability slightly relative to the measured stock; a stress-relief anneal after machining recovers it if a core ever runs near the knee.

3 Method

Geometry. Each point is a full-assembly STEP export from the parametrised Onshape document (5 configuration parameters), imported, meshed and solved in 3D by the Gmsh/GetDP pipeline (magnetic vector potential, edge elements, $\sim 5.9 \times 10^5$ unknowns). The chamber is fixed, so the channel region is taken from its CAD-confirmed dimensions rather than re-detected per run.

Linear iron, and why that is enough. Every sweep point uses linear iron ($\mu_r = 1500$), about 2.5 min per solve against ∼70 min nonlinear. Checked, not assumed: on the baseline geometry the linear solve reads +1.2% on the exit field relative to the committed nonlinear (A36-proxy) solve at equal current. A second check re-solved the highest-core-flux geometry (0.750 in core, 0.875 in spacer) with nonlinear iron. Its half-current stage converged (9 Picard iterations), but the full-current stage stalled at a relative residual of 0.11 instead of reaching 10^{-4}, so *it is not a validated solution.* Taken for what it is, its field agrees with the linear solve to 0.3% on the exit field and 0.5% in the outer cores, and puts the inner-core shaft flux density 3.4% *lower* (the linear figures are conservative there). The one sizeable disagreement is the 99.9th-percentile iron field (13% higher), i.e. a small number of high-field corner elements, which is plausibly where the Picard iteration is stuck. A finer ramp (nine stages, 50–100%) was then tried: the 50%, 60% and 70% stages converged (9, 16 and 35 iterations), but Picard contraction fell with every stage (about 2.3×, 1.7×, 1.2× per iteration) and at 75% dropped to about 1.1×, projecting past the 60-iteration cap, while solver memory grew to 13 GB. It was terminated. The difficulty rises with current level rather than step size, so a converged full-current check needs a Newton iteration, not a finer ramp. Because the model is linear, each geometry needs one solve: the field scales with ampere-turns, so the A-turns for exactly 300 G at the exit follow by scaling, and so do the iron flux densities.

Excitation. Inner and outer A-turns are held at the committed ratio (338 : 215 per coil, the field-direction optimum from the earlier current-ratio sweep) and scaled together to put the channel-averaged B_r at the exit station at 300 G. Sweeping the ratio per geometry is a cheap follow-up (two basis solves each) if the shape results below suggest it.

Metrics. The channel is split into 20 axial stations from anode to exit; B_r and B_{axial} are volume-element averages over each station. Radial purity is the mean of $B_r/|B|$. Core flux density is the mean axial B across the shaft cross-section (inside 90% of the core radius), taken at its maximum along the shaft, since leakage flux joins the core along its length.

Windings. `sizing/winding_design.py` turns A-turns into a winding. Wire dimensions are NEMA MW 1000 *maximum* overall diameters for heavy-build film-insulated wire (`sizing/wire_data.py`), not estimates, less a 0.15 mm radial allowance for ground insulation over the bare core. Both coils are wound full, each on its own constant-current channel, so no turn count has to be contrived to hit the A-turn ratio. Turn counts are reported under two packing models — hexagonal nesting (best case) and square packing (conservative) — because which one a hand-wound coil achieves is genuinely open; the tables here use square. Resistance uses the radius the copper actually occupies, not the window's. It reproduces the committed 280/336-turn AWG 20 design exactly.

Thermal. `thermal/radiation_balance.py`: vacuum, radiation from the outer envelope only (taken from each geometry's own mesh), emissivity 0.30, coils bonded to their cores. Temperatures feed back into copper resistance.

4 Results

4.1 Field in the circuit

Fig. 3 shows the flux path at both ends of the sweep: up the inner core, radially across the channel from the pole cap to the top-plate bore, down the outer cores and back through the bottom plate. The field in the channel is set by that exit gap, which is identical in every configuration, so the two maps differ mainly behind the channel. With a long spacer, the steel spacer sits just 4.8 mm from the inner core, across the inner winding, and offers flux a path back to the bottom plate that bypasses the channel (this is what loads the core; Sec. 4.4). With no spacer, the bottom plate sits 2.4 mm behind the anode face instead.

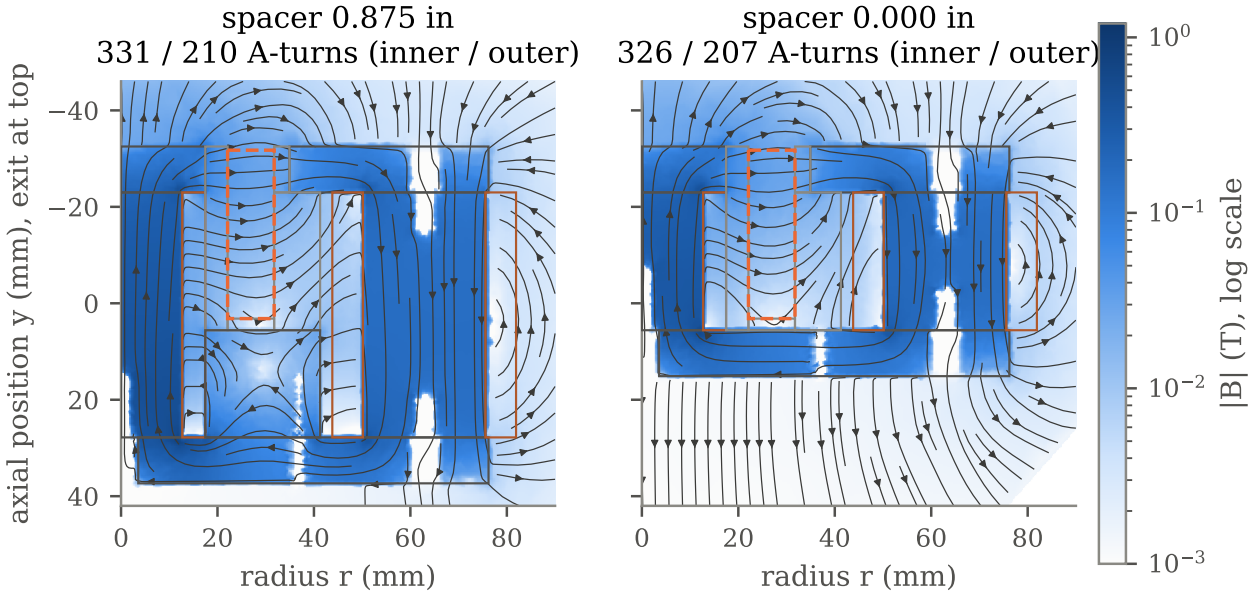


Figure 3: $|B|$ (log scale) and field lines in the plane through the axis and two opposite outer cores, at each geometry's own 300 G excitation, 1.000 in core. Left: longest spacer swept (0.875 in). Right: no spacer. Outlines: iron, windings, chamber; dashed orange: plasma channel.

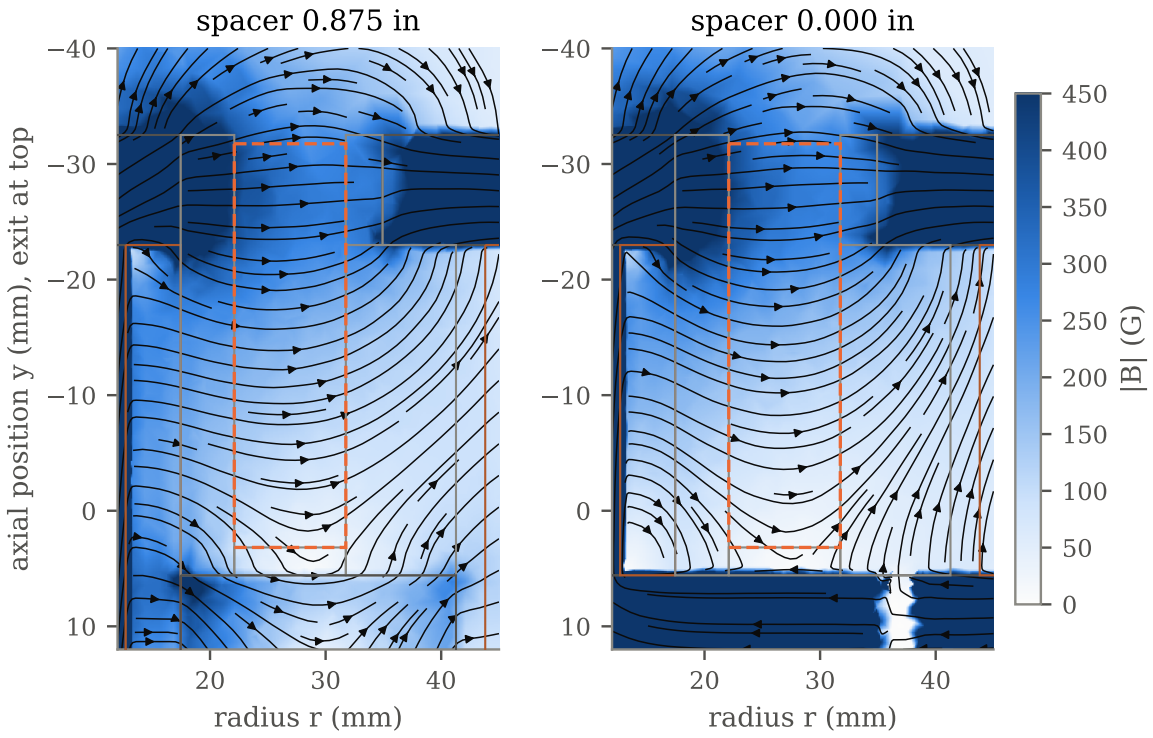


Figure 4: Close-up of the channel (dashed), linear colour scale in gauss, same two geometries. The field crosses the channel radially from the inner pole cap to the top-plate bore and weakens toward the anode.

4.2 Channel field shape

Every configuration keeps the same basic lens shape: B_r rises from the anode to a peak of about 317 G roughly 4 mm inside the exit plane, and is 300 G at the exit by construction (Fig. 5). The spacer only moves the anode half of the profile. Shorter spacers pull the anode-end field down (43 G at 0.875 in to 22 G at 0, 1.000 in core), which moves it toward the target lens and away from the anode, and they reduce the axial component there. The trade is a slightly less radial field overall: radial purity falls from 97.0% to 95.9% (Table 3). A thinner core shifts the anode field by only a few gauss. All configurations still overshoot the idealised target in the middle of the channel; that is a property of this pole geometry, not of the spacer.

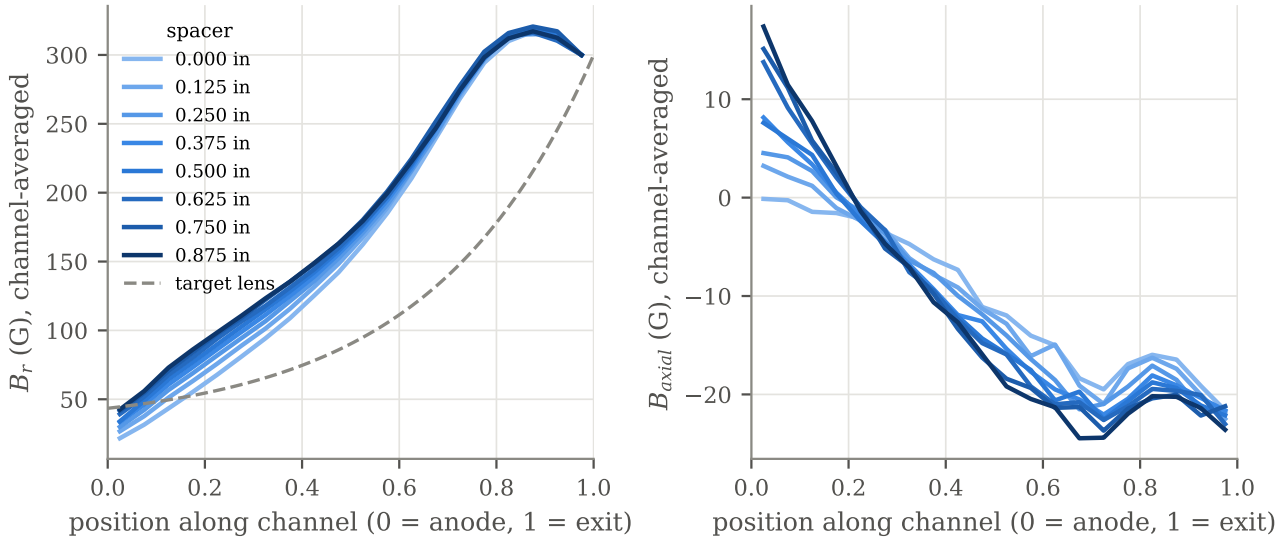


Figure 5: Channel-averaged radial (left) and axial (right) field from anode to exit, every spacer height, 1.000 in core, each at its own 300 G excitation. Dashed grey: the target magnetic-lens profile used by `field_quality.py`.

4.3 Ampere-turns needed

The A-turns needed for 300 G barely move (Fig. 6). Removing the spacer entirely changes the requirement by -1.3% at the 1.000 in core, comparable to the ±1% scatter between neighbouring points, which is mesh noise (each geometry is re-meshed from scratch). The one systematic trend is the core: a 0.750 in core needs about 2% more A-turns than a 1.000 in one, as more of the inner coil's flux leaks past the thinner core. Growing the outer coils to 1.625 in OD (one check point, 1.5 in `emag_height`) changes the requirement by -0.4%, i.e. not at all: outer-coil OD is purely a winding-room choice.

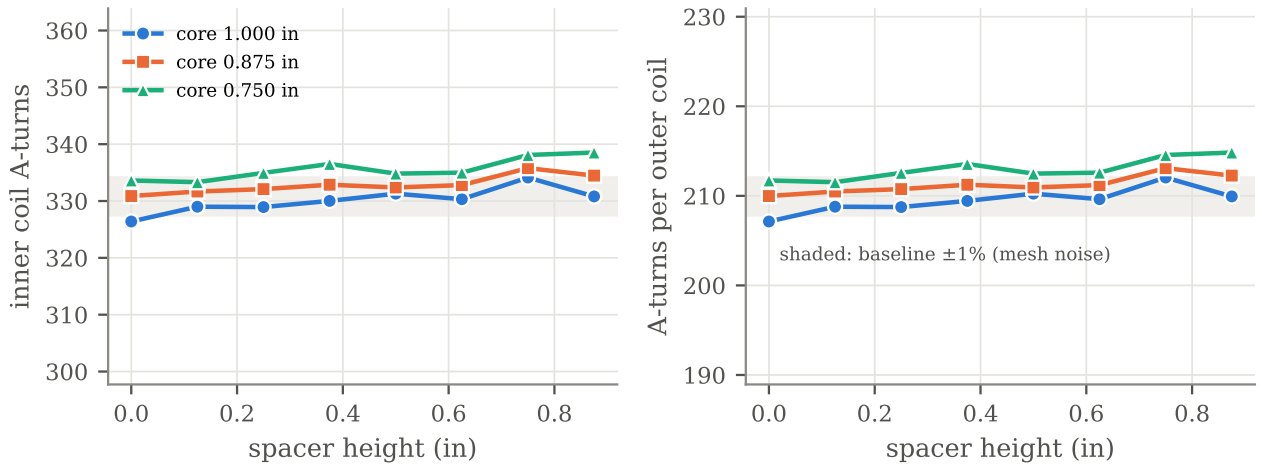


Figure 6: A-turns for 300 G at the channel exit, inner coil and each outer coil, and the resulting field per inner ampere-turn.

4.4 Core flux density

Every core stays far below the knee (Fig. 7). The inner core's shaft flux density runs 0.34–0.46 T at 1.000 in, 0.45–0.56 T at 0.875 in and 0.61–0.74 T at 0.750 in, scaling roughly with $1/d^2$ as it must for a fixed flux. The outer cores sit at 0.17–0.19 T throughout. Counter-intuitively, core flux is highest at the *longest* spacer. The

steel spacer wraps the inner coil and gives flux a return path through the bottom plate, so the lower part of the core carries that leakage on top of the useful channel flux. Shortening the spacer removes it. The worst case in the sweep, 0.74 T, is about half the 1.3–1.5 T knee; on the measured 1020 curve the cores stay on the steep, near-linear part of the curve everywhere swept.

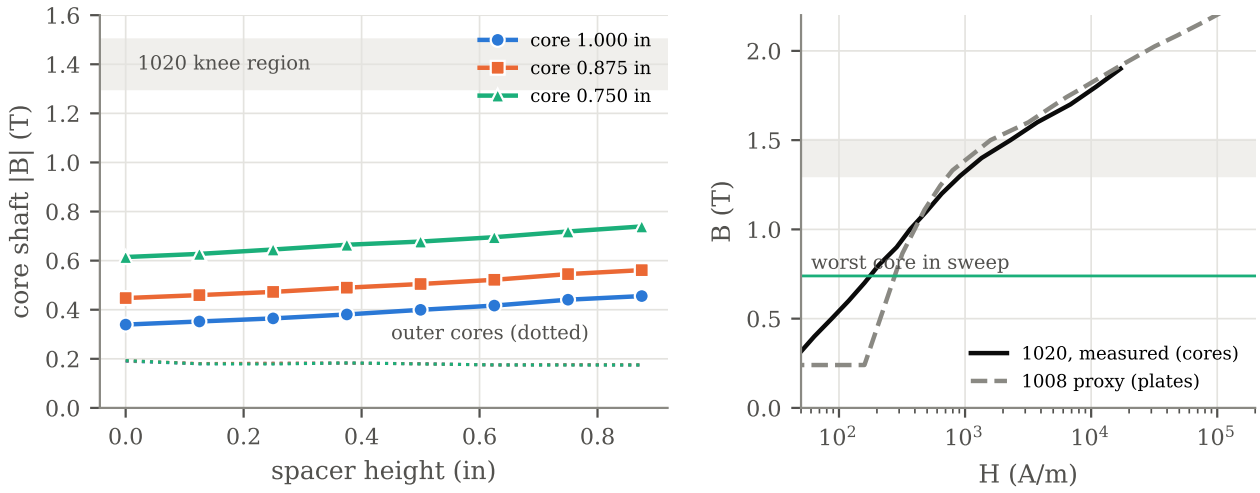


Figure 7: Left: highest shaft-averaged flux density along each core at 300 G, per inner-core diameter (solid: inner core; dotted: outer cores). Right: the measured 1020 curve used for the cores and the 1008 proxy used for the plates, with the worst swept core marked. Shaded: the knee.

4.5 Coil geometries and windings

Table 2 lists every coil geometry. Tables 3 and 4 give the magnetics and the AWG 20 winding for each; Appendix A has every gauge.

Three things stand out in Fig. 8:

- **Current and power track winding room, not the field.** The A-turns are almost constant, so current is inversely proportional to how many turns fit. The inner window is (core-to-chamber build) × `emag_height`, and both a short spacer and a thick core shrink it.
- **String voltage is nearly constant** at 4.0–4.5 V hot across every AWG 20 design. For a fixed gauge $V = IR = (NI)\,\rho\,L_{\text{turn}}/A_{\text{wire}}$: it depends only on the A-turns and the mean turn length, not on how many turns fit. Since the A-turns hardly change, neither does the voltage, and power (VI) follows current.
- **A thinner core offsets a shorter spacer.** Each step from 1.000 to 0.875 to 0.750 in roughly offsets 0.4 in of spacer. No spacer with a 0.750 in core (261/166 turns, 1.28 A, 5.4 W, 69°C) matches today's design (260/165 turns, 1.27 A, 5.1 W, 61°C) almost exactly.

Every configuration is thermally comfortable: the hottest coil is 95°C (1.000 in core, no spacer), against a 200°C wire-insulation limit.

spacer	emag_height	inner coil				outer coil (each)		
(in)	(in)	ID (in)	OD (in)	build (mm)	window (mm^2)	ID (in)	OD (in)	window (mm^2)
0.000	1.125	1.000	1.375	4.61	132	1.000	1.500	177
0.125	1.250	1.000	1.375	4.61	146	1.000	1.500	197
0.250†	1.375	1.000	1.375	4.61	161	1.000	1.500	217
0.375†	1.500	1.000	1.375	4.61	176	1.000	1.500	236
0.500†	1.625	1.000	1.375	4.61	190	1.000	1.500	256
0.625	1.750	1.000	1.375	4.61	205	1.000	1.500	276
0.750	1.875	1.000	1.375	4.61	220	1.000	1.500	295
0.875	2.000	1.000	1.375	4.61	234	1.000	1.500	315
0.000	1.125	0.875	1.375	6.20	177	1.000	1.500	177
0.125	1.250	0.875	1.375	6.20	197	1.000	1.500	197
0.250†	1.375	0.875	1.375	6.20	217	1.000	1.500	217
0.375†	1.500	0.875	1.375	6.20	236	1.000	1.500	236
0.500†	1.625	0.875	1.375	6.20	256	1.000	1.500	256
0.625	1.750	0.875	1.375	6.20	276	1.000	1.500	276
0.750	1.875	0.875	1.375	6.20	295	1.000	1.500	295
0.875	2.000	0.875	1.375	6.20	315	1.000	1.500	315
0.000	1.125	0.750	1.375	7.79	223	1.000	1.500	177
0.125	1.250	0.750	1.375	7.79	247	1.000	1.500	197
0.250†	1.375	0.750	1.375	7.79	272	1.000	1.500	217
0.375†	1.500	0.750	1.375	7.79	297	1.000	1.500	236
0.500†	1.625	0.750	1.375	7.79	321	1.000	1.500	256
0.625	1.750	0.750	1.375	7.79	346	1.000	1.500	276
0.750	1.875	0.750	1.375	7.79	371	1.000	1.500	295
0.875	2.000	0.750	1.375	7.79	396	1.000	1.500	315

Table 2: Coil geometry of every swept configuration. Inner coil ID = core diameter; OD is fixed at 1.375 in by the chamber bore. Build and window are the space available *to the winding*, i.e. net of a 0.15 mm radial allowance for ground insulation over the bare core — so the inner build reads 4.61 mm against the 4.76 mm of bare geometry. † spacer at a stocked A36 thickness.

spacer	core	A-turns for 300 G		B_r peak	B_r anode	radial	shaft $\lvert B\rvert$ (T)		iron
(in)	(in)	inner	outer ea.	(G)	(G)	purity (%)	inner	outer	p99.9 (T)
0.000	1.000	326	207	317	22	95.9	0.34	0.19	0.42
0.125	1.000	329	209	318	27	96.4	0.35	0.18	0.44
0.250†	1.000	329	209	315	30	96.6	0.36	0.18	0.45
0.375†	1.000	330	209	318	33	96.8	0.38	0.18	0.47
0.500†	1.000	331	210	320	34	96.9	0.40	0.18	0.49
0.625	1.000	330	210	316	39	96.9	0.42	0.17	0.50
0.750	1.000	334	212	321	42	97.0	0.44	0.18	0.53
0.875	1.000	331	210	317	43	97.0	0.46	0.17	0.55
0.000	0.875	331	210	317	20	95.5	0.45	0.19	0.52
0.125	0.875	332	210	317	24	96.1	0.46	0.18	0.54
0.250†	0.875	332	211	317	27	96.3	0.47	0.18	0.55
0.375†	0.875	333	211	316	30	96.5	0.49	0.18	0.56
0.500†	0.875	332	211	318	30	96.7	0.50	0.18	0.58
0.625	0.875	333	211	317	35	96.8	0.52	0.17	0.60
0.750	0.875	336	213	319	37	96.9	0.55	0.18	0.62
0.875	0.875	334	212	318	38	96.9	0.56	0.18	0.63
0.000	0.750	334	212	317	19	94.9	0.61	0.19	0.68
0.125	0.750	333	212	318	23	95.6	0.63	0.18	0.70
0.250†	0.750	335	213	318	25	95.9	0.65	0.18	0.72
0.375†	0.750	337	214	319	27	96.1	0.66	0.18	0.73
0.500†	0.750	335	212	317	28	96.3	0.68	0.18	0.74
0.625	0.750	335	213	317	32	96.5	0.70	0.18	0.76
0.750	0.750	338	215	317	33	96.6	0.72	0.17	0.79
0.875	0.750	339	215	318	34	96.7	0.74	0.17	0.80

Table 3: Magnetics per configuration, at each one's 300 G excitation (linear iron).

spacer	core	inner		outer (each)		current (A)		V_{hot} (V)		P_{hot}	T_{coil}	wire
(in)	(in)	turns	layers	turns	layers	inner	outer	inner	outer	(W)	(°C)	(m)
0.000	1.000	160	5	192	6	2.04	1.08	1.20	3.13	5.8	72	90
0.125	1.000	175	5	210	6	1.88	0.99	1.20	3.12	5.4	68	99
0.250†	1.000	195	5	234	6	1.69	0.89	1.18	3.07	4.7	63	110
0.375†	1.000	210	5	252	6	1.57	0.83	1.17	3.05	4.4	59	118
0.500†	1.000	230	5	276	6	1.44	0.76	1.16	3.03	4.0	56	129
0.625	1.000	245	5	294	6	1.35	0.71	1.15	2.99	3.7	54	138
0.750	1.000	265	5	318	6	1.26	0.67	1.15	3.01	3.5	52	149
0.875	1.000	280	5	336	6	1.18	0.62	1.13	2.95	3.2	49	158
0.000	0.875	192	6	192	6	1.72	1.09	1.12	3.15	5.4	69	92
0.125	0.875	210	6	210	6	1.58	1.00	1.10	3.11	4.9	64	100
0.250†	0.875	234	6	234	6	1.42	0.90	1.09	3.07	4.3	60	112
0.375†	0.875	252	6	252	6	1.32	0.84	1.08	3.05	4.0	57	120
0.500†	0.875	276	6	276	6	1.20	0.76	1.07	3.01	3.6	54	132
0.625	0.875	294	6	294	6	1.13	0.72	1.06	2.99	3.3	51	140
0.750	0.875	318	6	318	6	1.06	0.67	1.06	3.00	3.1	50	152
0.875	0.875	336	6	336	6	1.00	0.63	1.05	2.97	2.9	48	161
0.000	0.750	256	8	192	6	1.30	1.10	1.06	3.14	4.8	65	96
0.125	0.750	280	8	210	6	1.19	1.01	1.04	3.09	4.4	61	105
0.250†	0.750	312	8	234	6	1.07	0.91	1.03	3.07	3.9	57	117
0.375†	0.750	336	8	252	6	1.00	0.85	1.03	3.06	3.6	54	126
0.500†	0.750	368	8	276	6	0.91	0.77	1.01	3.01	3.2	51	138
0.625	0.750	392	8	294	6	0.85	0.72	1.01	2.99	3.0	49	147
0.750	0.750	424	8	318	6	0.80	0.67	1.01	3.00	2.8	47	159
0.875	0.750	448	8	336	6	0.76	0.64	1.00	2.99	2.7	46	168

Table 4: AWG 20 winding for every configuration (the committed gauge). Both coils are wound full on two independent constant-current channels, so the inner and outer currents differ; the four outer coils sit in series on the outer channel. Square packing. Other gauges: Appendix A.

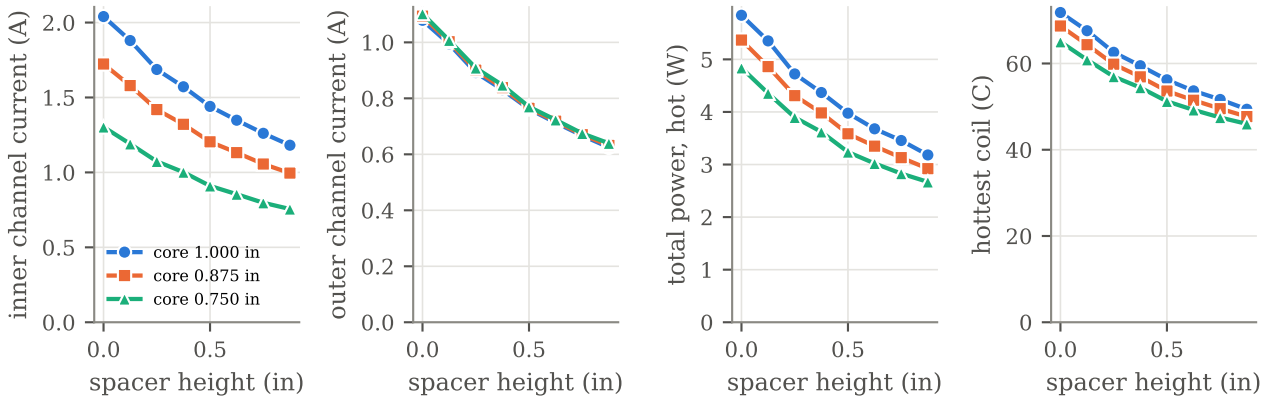


Figure 8: AWG 20: the two channel currents, total power at operating temperature, and hottest-coil temperature, per spacer height and core diameter.

5 Power-supply requirements

Nothing in the magnetics depends on the gauge: only the ampere-turn product reaches the field. The gauge sets how that product splits into current and voltage, and therefore which supply fits. Table 5 gives, for each gauge, the envelope over *every* swept geometry, so a supply chosen from it covers whichever spacer height is picked. Fig. 9 shows the same designs as operating points.

AWG	current (A)		max V, hot	max P	suggested rating,
	inner	outer	(V)	(W)	each channel
16	1.88–5.44	1.46–2.65	1.3	5.8	≥6.8 A, ≥5 V
17	1.41–4.73	1.05–1.84	1.6	5.6	≥5.9 A, ≥5 V
18	1.07–3.26	0.93–1.69	2.0	5.8	≥4.1 A, ≥5 V
19	0.95–2.91	0.69–1.26	2.5	5.9	≥3.6 A, ≥5 V
20	0.76–2.04	0.62–1.10	3.1	5.8	≥2.5 A, ≥6 V
21	0.60–1.87	0.48–0.86	4.0	6.2	≥2.3 A, ≥8 V
22	0.48–1.36	0.37–0.66	5.1	5.9	≥1.7 A, ≥10 V
23	0.36–1.06	0.30–0.53	6.4	5.9	≥1.3 A, ≥13 V
24	0.30–0.95	0.24–0.43	8.1	6.3	≥1.2 A, ≥16 V

Table 5: Supply envelope per wire gauge across all swept geometries, for two independent constant-current channels. "Hot" is at the steady-state coil temperature. Suggested rating: 1.25× the highest current on either channel (room for the linear-model and wire-tolerance uncertainty and a field margin above 300 G) and 2× the highest hot voltage (room for lead resistance and ramping into the inductance), floored at 5 V. A two-channel 30 V/3 A bench supply covers every AWG 20 design.

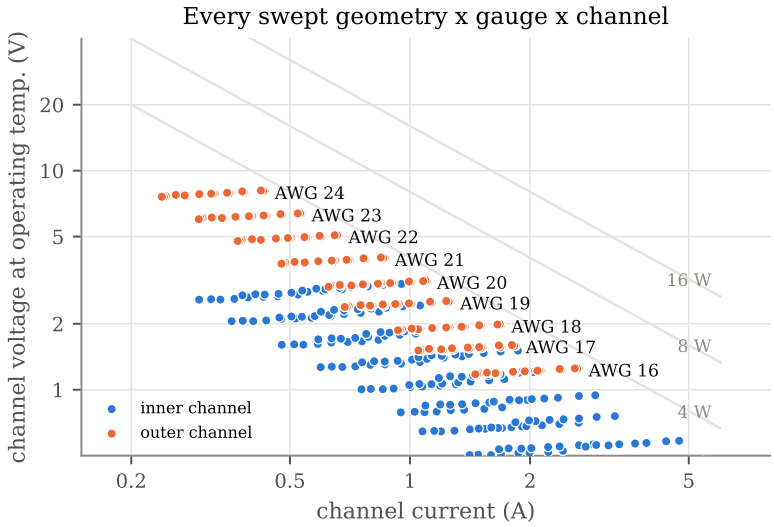


Figure 9: Every swept geometry in every gauge as a (current, voltage) operating point, one per channel, with lines of constant power. Gauge moves a design along a constant-power line; geometry moves it across the lines. The outer channel sits at lower current and higher voltage because its coils are wound full.

What the supply needs, independent of gauge:

- **Constant-current (CC) regulation.** The field follows current, not voltage; copper resistance rises about 0.4%/K as the coils warm, so a constant-voltage supply would let the field drift down by the same amount. Any bench supply with a CC mode works; the load is a plain resistor-inductor.
- **Two channels, not one.** The inner coil and the four outer coils (in series with each other) run on separate constant-current channels. One series string would work, but it forces the outer coils to be part-wound to whatever turn count gives the right ampere-turn ratio at the shared current, which costs 1.46× the dissipation and freezes that ratio in copper — a ratio that came from a linear solve with a proxy B-H curve. With two channels both coils are wound full and the ratio stays a dial. It also removes the turn-count risk: count the turns actually wound and set each channel to A-turns/N.
- **Inductance, and switching off.** Stored field energy at the operating point is 66 mJ, set by the field and therefore the same at every gauge. With two channels this does not collapse to a single string inductance: splitting it into two self-inductances plus a mutual term needs one solve per channel energised alone, which has not been done. What matters practically is unaffected — the time constant is of order 100 ms, so ramp the current over $\geq$1 s rather than stepping it, and put a flyback diode across each channel so an open circuit or a relay opening under current cannot generate a voltage spike. At 66 mJ the diode choice is trivial (any 1 A/100 V part); the inductance constrains ramp rate, not the diode. Ripple is heavily filtered by the same time constant and does not matter.
- **Polarity.** The inner winding is driven opposite to the outer windings (flux crosses the channel from the inner pole to the top plate). Wire the string so the inner coil's sense is reversed relative to the four outer coils, and check it with a gaussmeter at first power-up: wrong polarity collapses the channel field.
- **Floating output.** The coils sit next to the discharge; an isolated (floating) output keeps the magnet circuit independent of the discharge supply's reference.

6 Limitations and open items

- **Ground insulation is an allowance.** Wire dimensions are now NEMA MW 1000 maxima rather than estimates, but the 0.15 mm radial allowance for ground insulation over the bare core is a build assumption, and it is the input most likely to cost the inner coil a layer (0.152 mm of unused build remains at AWG 20). Which packing model the wound coils achieve is also open; both are reported and the tables use the conservative one.
- **Linear iron.** Each point is a linear solve. It is checked against a converged nonlinear solve only at the baseline geometry; the worst-case nonlinear check did not converge (Sec. 3), though its unconverged field

agrees with the linear one in the channel and cores. The plates' B-H curve is a proxy (1008 for A36). The circuit is gap-dominated, so this mostly matters for the core saturation margin, which is judged against the measured 1020 curve.

- **Fixed inner/outer A-turn ratio.** Re-optimising the ratio per geometry could recover some field shape at short spacers. The two-channel supply makes this adjustable on the bench rather than fixed at winding time, so it is now a commissioning question rather than a design one.
- **The 300 G target is not derived** from the plasma requirements (the top-level sizing script assumes 150 G). The results scale linearly: halving the target halves every A-turn and current figure and quarters every power.
- **Magnets-only thermal.** No plasma heat load; emissivity 0.30 assumed (`thermal/README.md` has the sensitivity).

A Every gauge, every geometry

spacer (in)	core (in)	AWG	turns in/out	current (A) inner	current (A) outer	V_{hot} (V) inner	V_{hot} (V) outer	P_{hot} (W)	T_{coil} (°C)	J_{inner} (A/mm^2)
0.000	1.000	16	60/80	5.44	2.59	0.47	1.25	5.8	72	4.2
		17	69/115	4.73	1.80	0.58	1.60	5.6	71	4.5
		18	100/125	3.26	1.66	0.76	1.99	5.8	71	4.0
		19	112/168	2.91	1.23	0.94	2.54	5.9	72	4.5
		20	160/192	2.04	1.08	1.20	3.13	5.8	72	3.9
		21	175/245	1.87	0.85	1.51	4.01	6.2	74	4.5
		22	240/320	1.36	0.65	1.92	5.07	5.9	72	4.2
		23	308/396	1.06	0.52	2.42	6.38	5.9	72	4.1
		24	343/490	0.95	0.42	3.04	8.12	6.3	75	4.6
0.125	1.000	16	66/88	4.98	2.37	0.47	1.24	5.3	67	3.8
		17	75/125	4.39	1.67	0.58	1.59	5.2	67	4.2
		18	112/140	2.94	1.49	0.75	1.97	5.1	66	3.6
		19	124/186	2.65	1.12	0.94	2.52	5.3	68	4.1
		20	175/210	1.88	0.99	1.20	3.12	5.4	68	3.6
		21	195/273	1.69	0.76	1.49	3.98	5.6	69	4.1
		22	264/352	1.25	0.59	1.90	5.04	5.4	68	3.8
		23	343/441	0.96	0.47	2.40	6.34	5.3	67	3.7
		24	385/550	0.85	0.38	3.01	8.04	5.6	70	4.2
0.250†	1.000	16	75/100	4.39	2.09	0.46	1.22	4.6	62	3.4
		17	84/140	3.92	1.49	0.57	1.57	4.6	62	3.8
		18	124/155	2.65	1.35	0.74	1.94	4.6	62	3.2
		19	140/210	2.35	0.99	0.92	2.48	4.6	62	3.6
		20	195/234	1.69	0.89	1.18	3.07	4.7	63	3.2
		21	215/301	1.53	0.69	1.47	3.92	5.0	64	3.7
		22	288/384	1.14	0.54	1.88	4.97	4.8	63	3.5
		23	378/486	0.87	0.43	2.37	6.24	4.7	63	3.4
		24	420/600	0.78	0.35	2.96	7.93	5.1	65	3.8
0.375†	1.000	16	81/108	4.07	1.94	0.46	1.21	4.2	59	3.1
		17	90/150	3.67	1.40	0.57	1.56	4.3	59	3.5
		18	136/170	2.43	1.23	0.73	1.92	4.1	58	2.9
		19	152/228	2.17	0.92	0.91	2.46	4.2	59	3.3
		20	210/252	1.57	0.83	1.17	3.05	4.4	59	3.0
		21	235/329	1.40	0.64	1.46	3.89	4.5	61	3.4
		22	318/424	1.04	0.49	1.86	4.93	4.4	59	3.2
		23	413/531	0.80	0.39	2.35	6.19	4.3	59	3.1
		24	462/660	0.71	0.32	2.93	7.85	4.6	61	3.5
0.500†	1.000	16	87/116	3.81	1.81	0.46	1.21	3.9	56	2.9
		17	99/165	3.35	1.27	0.56	1.55	3.9	56	3.2
		18	148/185	2.24	1.14	0.73	1.91	3.8	55	2.7
		19	164/246	2.02	0.85	0.91	2.45	3.9	56	3.1

spacer (in)	core (in)	AWG	turns in/out	current (A) inner	current (A) outer	V_{hot} (V) inner	V_{hot} (V) outer	P_{hot} (W)	T_{coil} (°C)	J_{inner} (A/mm^2)
		20	230/276	1.44	0.76	1.16	3.03	4.0	56	2.8
		21	255/357	1.30	0.59	1.45	3.86	4.2	57	3.2
		22	342/456	0.97	0.46	1.85	4.90	4.0	57	3.0
		23	448/576	0.74	0.37	2.33	6.15	4.0	56	2.9
		24	497/710	0.67	0.30	2.92	7.81	4.3	58	3.3
0.625	1.000	16	96/128	3.44	1.64	0.45	1.19	3.5	52	2.6
		17	105/175	3.15	1.20	0.56	1.53	3.6	53	3.0
		18	160/200	2.06	1.05	0.72	1.89	3.5	52	2.5
		19	176/264	1.88	0.79	0.90	2.42	3.6	53	2.9
		20	245/294	1.35	0.71	1.15	2.99	3.7	54	2.6
		21	275/385	1.20	0.54	1.43	3.81	3.8	54	2.9
		22	372/496	0.89	0.42	1.82	4.84	3.7	54	2.7
		23	483/621	0.68	0.34	2.30	6.08	3.6	53	2.6
		24	539/770	0.61	0.27	2.87	7.70	3.9	55	3.0
0.750	1.000	16	102/136	3.28	1.56	0.45	1.20	3.3	51	2.5
		17	114/190	2.93	1.12	0.56	1.54	3.4	51	2.8
		18	168/210	1.99	1.01	0.72	1.90	3.4	51	2.4
		19	188/282	1.78	0.75	0.90	2.43	3.4	52	2.7
		20	265/318	1.26	0.67	1.15	3.01	3.5	52	2.4
		21	295/413	1.13	0.51	1.44	3.83	3.6	53	2.8
		22	396/528	0.84	0.40	1.83	4.87	3.5	52	2.6
		23	518/666	0.65	0.32	2.31	6.11	3.4	52	2.5
		24	574/820	0.58	0.26	2.89	7.75	3.7	53	2.8
0.875	1.000	16	108/144	3.06	1.46	0.44	1.18	3.1	49	2.3
		17	120/200	2.76	1.05	0.55	1.51	3.1	49	2.6
		18	180/225	1.84	0.93	0.71	1.87	3.0	49	2.2
		19	204/306	1.62	0.69	0.88	2.39	3.1	49	2.5
		20	280/336	1.18	0.62	1.13	2.95	3.2	49	2.3
		21	315/441	1.05	0.48	1.41	3.76	3.3	50	2.6
		22	426/568	0.78	0.37	1.80	4.77	3.2	49	2.4
		23	553/711	0.60	0.30	2.27	6.00	3.1	49	2.3
		24	616/880	0.54	0.24	2.83	7.60	3.3	50	2.6
0.000	0.875	16	80/80	4.14	2.62	0.44	1.25	5.1	67	3.2
		17	115/115	2.88	1.83	0.56	1.58	4.5	63	2.8
		18	125/125	2.65	1.68	0.71	1.99	5.2	68	3.2
		19	168/168	1.97	1.25	0.90	2.53	4.9	66	3.0
		20	192/192	1.72	1.09	1.12	3.15	5.4	69	3.3
		21	245/245	1.35	0.86	1.42	4.00	5.3	69	3.3
		22	320/320	1.03	0.66	1.80	5.07	5.2	68	3.2
		23	396/396	0.84	0.53	2.27	6.39	5.3	68	3.2
		24	490/490	0.68	0.43	2.87	8.08	5.4	69	3.3
0.125	0.875	16	88/88	3.77	2.39	0.44	1.24	4.6	63	2.9
		17	125/125	2.65	1.68	0.56	1.57	4.1	59	2.6
		18	140/140	2.37	1.50	0.70	1.96	4.6	63	2.9
		19	186/186	1.78	1.13	0.89	2.50	4.4	61	2.7
		20	210/210	1.58	1.00	1.10	3.11	4.9	64	3.0
		21	273/273	1.21	0.77	1.40	3.94	4.7	64	3.0
		22	352/352	0.94	0.60	1.78	5.01	4.7	63	2.9
		23	441/441	0.75	0.48	2.24	6.31	4.7	63	2.9
		24	550/550	0.60	0.38	2.83	7.96	4.8	64	2.9
0.250[†]	0.875	16	100/100	3.32	2.11	0.43	1.22	4.0	58	2.5
		17	140/140	2.37	1.51	0.55	1.55	3.6	55	2.3
		18	155/155	2.14	1.36	0.69	1.94	4.1	58	2.6
		19	210/210	1.58	1.00	0.87	2.46	3.9	57	2.4
		20	234/234	1.42	0.90	1.09	3.07	4.3	60	2.7
		21	301/301	1.10	0.70	1.38	3.90	4.3	59	2.7

spacer (in)	core (in)	AWG	turns in/out	current (A) inner	current (A) outer	V_{hot} (V) inner	V_{hot} (V) outer	P_{hot} (W)	T_{coil} (°C)	J_{inner} (A/mm^2)
		22	384/384	0.86	0.55	1.76	4.96	4.2	59	2.7
		23	486/486	0.68	0.43	2.21	6.23	4.2	59	2.6
		24	600/600	0.55	0.35	2.80	7.88	4.3	60	2.7
0.375†	0.875	16	108/108	3.08	1.96	0.43	1.21	3.7	55	2.4
		17	150/150	2.22	1.41	0.55	1.54	3.4	53	2.1
		18	170/170	1.96	1.24	0.68	1.92	3.7	55	2.4
		19	228/228	1.46	0.93	0.87	2.44	3.5	54	2.2
		20	252/252	1.32	0.84	1.08	3.05	4.0	57	2.5
		21	329/329	1.01	0.64	1.37	3.86	3.9	56	2.5
		22	424/424	0.79	0.50	1.74	4.91	3.8	56	2.4
		23	531/531	0.63	0.40	2.19	6.18	3.8	56	2.4
		24	660/660	0.50	0.32	2.77	7.80	3.9	56	2.5
0.500†	0.875	16	116/116	2.87	1.82	0.42	1.20	3.4	52	2.2
		17	165/165	2.01	1.28	0.54	1.52	3.0	50	1.9
		18	185/185	1.80	1.14	0.67	1.90	3.4	52	2.2
		19	246/246	1.35	0.86	0.86	2.42	3.2	51	2.1
		20	276/276	1.20	0.76	1.07	3.01	3.6	54	2.3
		21	357/357	0.93	0.59	1.35	3.82	3.5	53	2.3
		22	456/456	0.73	0.46	1.72	4.86	3.5	53	2.2
		23	576/576	0.58	0.37	2.17	6.11	3.5	53	2.2
		24	710/710	0.47	0.30	2.74	7.72	3.6	54	2.3
0.625	0.875	16	128/128	2.60	1.65	0.42	1.19	3.1	50	2.0
		17	175/175	1.90	1.21	0.54	1.52	2.9	48	1.8
		18	200/200	1.66	1.06	0.67	1.89	3.1	50	2.0
		19	264/264	1.26	0.80	0.85	2.40	3.0	49	1.9
		20	294/294	1.13	0.72	1.06	2.99	3.3	51	2.2
		21	385/385	0.86	0.55	1.35	3.79	3.2	51	2.1
		22	496/496	0.67	0.43	1.71	4.82	3.2	50	2.1
		23	621/621	0.54	0.34	2.15	6.07	3.2	51	2.1
		24	770/770	0.43	0.27	2.72	7.66	3.3	51	2.1
0.750	0.875	16	136/136	2.47	1.57	0.42	1.19	2.9	48	1.9
		17	190/190	1.77	1.12	0.54	1.52	2.7	46	1.7
		18	210/210	1.60	1.01	0.67	1.90	3.0	49	1.9
		19	282/282	1.19	0.76	0.86	2.41	2.8	48	1.8
		20	318/318	1.06	0.67	1.06	3.00	3.1	50	2.0
		21	413/413	0.81	0.52	1.35	3.81	3.1	49	2.0
		22	528/528	0.64	0.40	1.72	4.84	3.0	49	2.0
		23	666/666	0.50	0.32	2.16	6.09	3.0	49	1.9
		24	820/820	0.41	0.26	2.73	7.69	3.1	49	2.0
0.875	0.875	16	144/144	2.32	1.47	0.42	1.18	2.7	46	1.8
		17	200/200	1.67	1.06	0.53	1.51	2.5	45	1.6
		18	225/225	1.49	0.94	0.66	1.88	2.8	47	1.8
		19	306/306	1.09	0.69	0.85	2.39	2.6	45	1.7
		20	336/336	1.00	0.63	1.05	2.97	2.9	48	1.9
		21	441/441	0.76	0.48	1.33	3.76	2.8	47	1.8
		22	568/568	0.59	0.37	1.70	4.79	2.8	47	1.8
		23	711/711	0.47	0.30	2.14	6.02	2.8	47	1.8
		24	880/880	0.38	0.24	2.70	7.60	2.9	47	1.9
0.000	0.750	16	100/80	3.34	2.65	0.42	1.25	4.7	64	2.6
		17	138/115	2.42	1.84	0.53	1.58	4.2	61	2.3
		18	175/125	1.91	1.69	0.68	1.98	4.7	64	2.3
		19	196/168	1.70	1.26	0.83	2.53	4.6	64	2.6
		20	256/192	1.30	1.10	1.06	3.14	4.8	65	2.5
		21	315/245	1.06	0.86	1.34	3.99	4.9	65	2.6
		22	400/320	0.83	0.66	1.69	5.07	4.8	65	2.6
		23	528/396	0.63	0.53	2.16	6.37	4.8	65	2.4

spacer (in)	core (in)	AWG	turns in/out	current (A) inner	current (A) outer	V_{hot} (V) inner	V_{hot} (V) outer	P_{hot} (W)	T_{coil} (°C)	J_{inner} (A/mm^2)
		24	637/490	0.52	0.43	2.72	8.06	4.9	66	2.6
0.125	0.750	16	110/88	3.03	2.40	0.41	1.23	4.2	60	2.3
		17	150/125	2.22	1.69	0.52	1.57	3.8	57	2.1
		18	196/140	1.70	1.51	0.67	1.95	4.1	59	2.1
		19	217/186	1.54	1.14	0.82	2.49	4.1	59	2.4
		20	280/210	1.19	1.01	1.04	3.09	4.4	61	2.3
		21	351/273	0.95	0.77	1.31	3.93	4.3	60	2.3
		22	440/352	0.76	0.60	1.66	4.99	4.3	60	2.3
		23	588/441	0.57	0.48	2.13	6.27	4.2	60	2.2
		24	715/550	0.47	0.38	2.67	7.92	4.3	60	2.3
0.250†	0.750	16	125/100	2.68	2.13	0.40	1.22	3.7	56	2.1
		17	168/140	1.99	1.52	0.52	1.55	3.4	54	1.9
		18	217/155	1.54	1.37	0.66	1.94	3.7	55	1.9
		19	245/210	1.37	1.01	0.81	2.47	3.6	55	2.1
		20	312/234	1.07	0.91	1.03	3.07	3.9	57	2.1
		21	387/301	0.87	0.71	1.31	3.90	3.9	57	2.1
		22	480/384	0.70	0.55	1.65	4.96	3.9	57	2.1
		23	648/486	0.52	0.44	2.11	6.23	3.8	56	2.0
		24	780/600	0.43	0.35	2.65	7.88	3.9	57	2.1
0.375†	0.750	16	135/108	2.49	1.98	0.40	1.22	3.4	53	1.9
		17	180/150	1.87	1.42	0.51	1.55	3.2	51	1.8
		18	238/170	1.41	1.26	0.66	1.93	3.4	53	1.7
		19	266/228	1.27	0.94	0.81	2.46	3.3	52	1.9
		20	336/252	1.00	0.85	1.03	3.06	3.6	54	1.9
		21	423/329	0.80	0.65	1.30	3.88	3.6	54	1.9
		22	530/424	0.63	0.50	1.64	4.93	3.5	54	2.0
		23	708/531	0.48	0.40	2.10	6.20	3.5	54	1.8
		24	858/660	0.39	0.32	2.64	7.83	3.6	54	1.9
0.500†	0.750	16	145/116	2.31	1.83	0.40	1.20	3.1	50	1.8
		17	198/165	1.69	1.29	0.51	1.53	2.8	48	1.6
		18	259/185	1.29	1.15	0.65	1.90	3.0	50	1.6
		19	287/246	1.17	0.86	0.79	2.42	3.0	50	1.8
		20	368/276	0.91	0.77	1.01	3.01	3.2	51	1.8
		21	459/357	0.73	0.60	1.28	3.82	3.2	51	1.8
		22	570/456	0.59	0.47	1.62	4.86	3.2	51	1.8
		23	768/576	0.44	0.37	2.07	6.11	3.2	51	1.7
		24	923/710	0.36	0.30	2.60	7.72	3.3	51	1.8
0.625	0.750	16	160/128	2.09	1.66	0.39	1.19	2.8	48	1.6
		17	210/175	1.60	1.21	0.50	1.52	2.6	47	1.5
		18	280/200	1.20	1.06	0.65	1.89	2.8	48	1.5
		19	308/264	1.09	0.81	0.79	2.41	2.8	48	1.7
		20	392/294	0.85	0.72	1.01	2.99	3.0	49	1.6
		21	495/385	0.68	0.55	1.27	3.79	3.0	49	1.6
		22	620/496	0.54	0.43	1.60	4.82	2.9	49	1.7
		23	828/621	0.40	0.34	2.06	6.07	2.9	48	1.6
		24	1001/770	0.33	0.28	2.58	7.66	3.0	49	1.6
0.750	0.750	16	170/136	1.99	1.58	0.40	1.19	2.7	46	1.5
		17	228/190	1.48	1.13	0.50	1.52	2.5	45	1.4
		18	294/210	1.15	1.02	0.65	1.90	2.7	46	1.4
		19	329/282	1.03	0.76	0.79	2.42	2.7	46	1.6
		20	424/318	0.80	0.67	1.01	3.00	2.8	47	1.5
		21	531/413	0.64	0.52	1.27	3.81	2.8	47	1.5
		22	660/528	0.51	0.41	1.61	4.85	2.8	47	1.6
		23	888/666	0.38	0.32	2.06	6.09	2.7	47	1.5
		24	1066/820	0.32	0.26	2.59	7.69	2.8	48	1.5
0.875	0.750	16	180/144	1.88	1.49	0.39	1.19	2.5	45	1.4

spacer (in)	core (in)	AWG	turns in/out	current (A) inner	current (A) outer	V_{hot} (V) inner	V_{hot} (V) outer	P_{hot} (W)	T_{coil} (°C)	J_{inner} (A/mm^2)
		17	240/200	1.41	1.07	0.50	1.52	2.3	44	1.4
		18	315/225	1.07	0.95	0.65	1.89	2.5	45	1.3
		19	357/306	0.95	0.70	0.79	2.41	2.4	44	1.5
		20	448/336	0.76	0.64	1.00	2.99	2.7	46	1.5
		21	567/441	0.60	0.49	1.27	3.79	2.6	46	1.5
		22	710/568	0.48	0.38	1.60	4.82	2.6	45	1.5
		23	948/711	0.36	0.30	2.05	6.06	2.6	45	1.4
		24	1144/880	0.30	0.24	2.58	7.66	2.6	46	1.4

B Reproducing this report

From the repository root:

```
cd gmsh_getdp/assembly
python emag_sweep.py --validate            # linear vs committed nonlinear, baseline
python emag_sweep.py                       # full grid, resumable (~1 h)
python emag_sweep.py --nonlinear h1.125_ci0.750_oo1.500
cd ../..
python report/render_3d.py
python report/make_report.py               # figures, tables, PDF (needs tools/tectonic)
```